Origami Axioms through GeoGebra

Ernest Scheiber*

Abstract

The origami folds corresponding to the axioms are available as GeoGebra tools. Some well known examples using the tools are presented.

1 Introduction

The origami folds corresponding to the seven *Huzita* axioms, [3], [6], [4], are given as GeoGebra tools. Set of rules to develop origami constructions are given in [1], [2], too.

GeoGebra is one of the leading mathematical software for learning and teaching, [7]. We suppose that the reader is familiarized with GeoGebra.

Our developed tools are available at `https://github.com/e-scheiber/origami.git`.

After presenting the tools, several examples are given. Each construction consists of a sequence of calling statements of the tools. Sometimes there is required to define and denote the intersection of two lines or to define a point on a line.

2 The origami GeoGebra tools

A tool is contained in a `.ggt` type file and is a GeoGebra procedure whose arguments are GeoGebra points. The first two arguments correspond to the bottom side of an origami square piece of paper. So, in $Hi[A, B, \ldots]$, $i \in \{1, \ldots, 7\}$, A and B denotes the left and the right side ends of the bottom line of a square.

*scheiber@unitbv.ro

Axiom 1

Given two points P_1 and P_2, there is a unique fold passing through both of them.

If E anf F are two points then $H1[A, B, E, F]$ draws the fold through the line EF, Fig. 1.

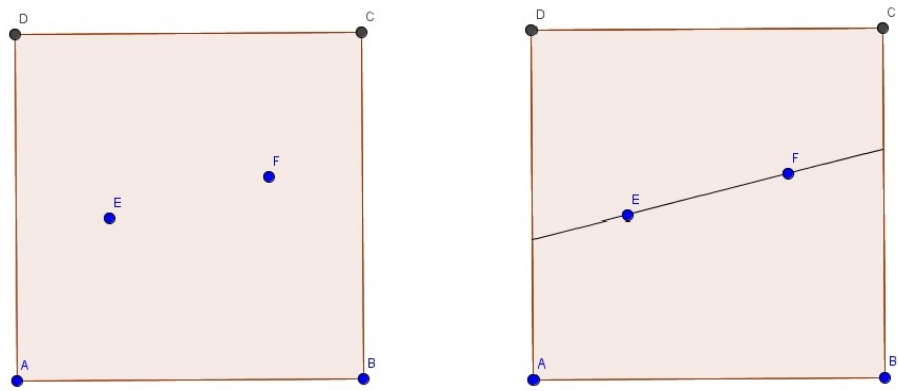

Fig. 1: Axiom 1.

The line $L(E, F)$ intersects the sides of the $ABCD$ origami square in the points $G \in AB, J \in BC, I \in CD, H \in AD$. For each of the intersection points there are computed the ratio between a segment determined by the point and the side of the square, e.g. on the AB line there is computed

$$q_s = \frac{x(G) - x(A)}{x(B) - x(A)} \in \mathbb{R}.$$

If the ratio belongs to $[0, 1]$ then the intersection point is on a side of the origami square. A weight is attached to the intersection points: 1 on AB, 2 on BC, 4 on CD, 8 on DA and 0 if the point in not on the side. The sum of the weights determine the intersection between the line and the square, e.g. if the sum is 6 then the intersection segment is defined by the intersections with the sides BC and CD and it is IJ. A special treatment should be given in that case when an intersection point is a corner of the square.

Axiom 2

Given two points P_1 and P_2, there is a unique fold placing P_1 onto P_2.

If E anf F are two points then $H2[A, B, E, F]$ draws the fold placing P_1 onto P_2, Fig. 2.

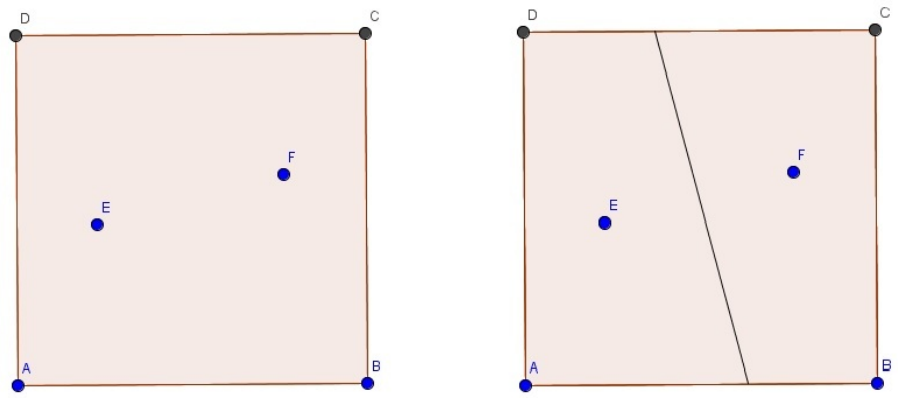

Fig. 2: Axiom 2.

Axiom 3

Given two lines L_1 and L_2 there is a fold placing L_1 onto L_2.

If $L_1(E, F)$ and $L_2(G, H)$ are the two lines defined by the points in the parentheses then $H3[A, B, E, F, G, H]$ draws the fold placing L_1 onto L_2, Fig. 3.

The programming problem is to find out if the lines are concurrent or parallel. Let be I their intersention point. If the lines are parallel then I is unknown for GeoGebra, (I =?). The fold line will be

```
If[(x(I) >= 0) || (x(I)  <  0),
   AngleBisector[If[I == E, F, E], I, If[I == G, H, G]],
   Line[Midpoint[E, G], Midpoint[F, H]]]
```

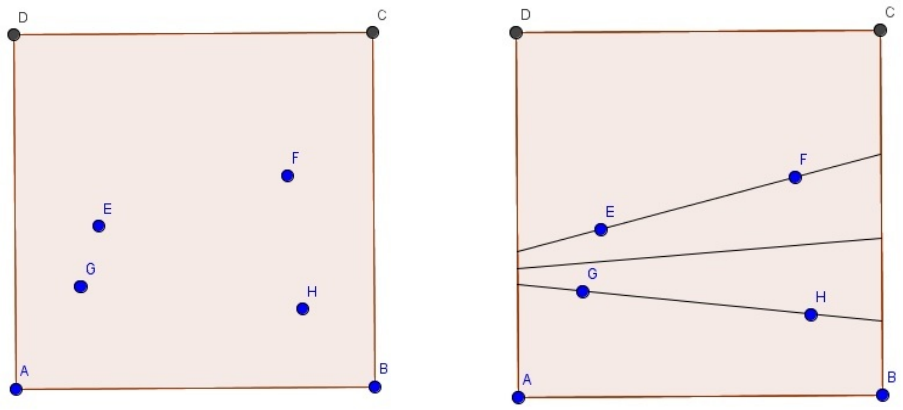

Fig. 3: Axiom 3.

Axiom 4

Given a point P and a line L, there is a unique fold perpendicular to L passing through P.

Let be the point E and the line $L(F, G)$, the tool $H4[A, B, E, F, G]$ draws the fold line passing through E and perpendicular to $L(F, G)$, Fig. 4.

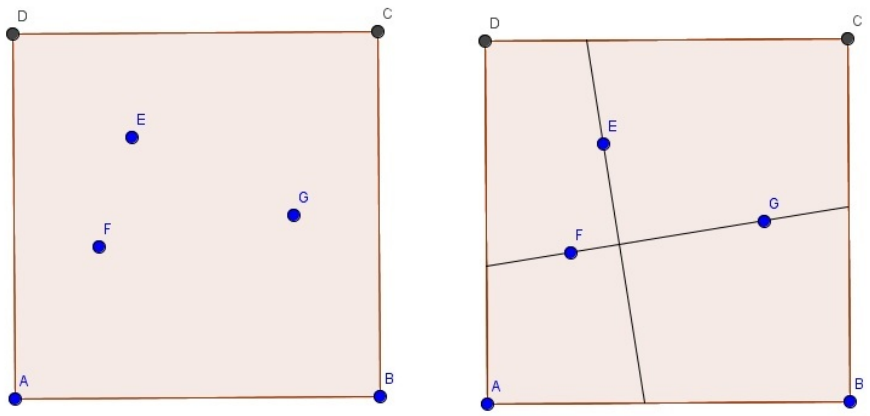

Fig. 4: Axiom 4.

Axiom 5

Given two points P_1 and P_2 and a line L then whenever possible there is a fold placing P_1 onto L and passing through P_2.

If $E,\ F$ are two points and the line $L(G,H)$ then the tool $H5[A,B,E,F,G,H]$ draws the fold passing through F and placing E onto the line $L(G,H)$, Fig. 5.

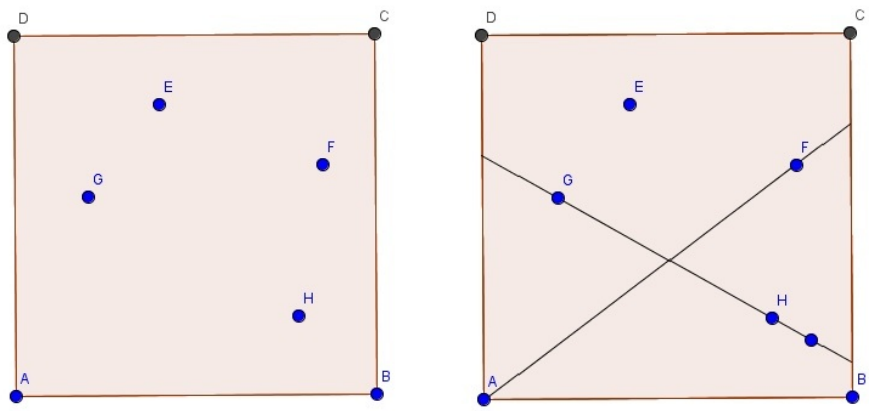

Fig. 5: Axiom 5.

The point without notation on the line $L(G,H)$ is the reflection of the point E about the fold line.

The construction involves the intersection of a line with a circle. Let $\mathcal{C}$ be the circle with center F through E and I,J the intersection of the circle with $L(G,H)$. Introducing

```
U=If[(x(I) >= 0) || (x(I)  <  0), I]
```

the fold line is the perpendicular bisector of the segment EU.

Because we have arbitrary selected one of the intersection points sometimes, in the calling statement, it requires to change the order of the points which define the line.

Axiom 6

Given two points P_1 and P_2 and two lines L_1 and L_2, there is a fold placing P_1 onto L_1 and P_2 onto L_2.

Let be the points E, F and the lines $L(G,H)$ and $L(I,J)$. The tool $H6[A,B,E,F,G,H,I,J]$ draws the fold placing E on $L(G,H)$ and F on $L(I,J)$, Fig. 6.

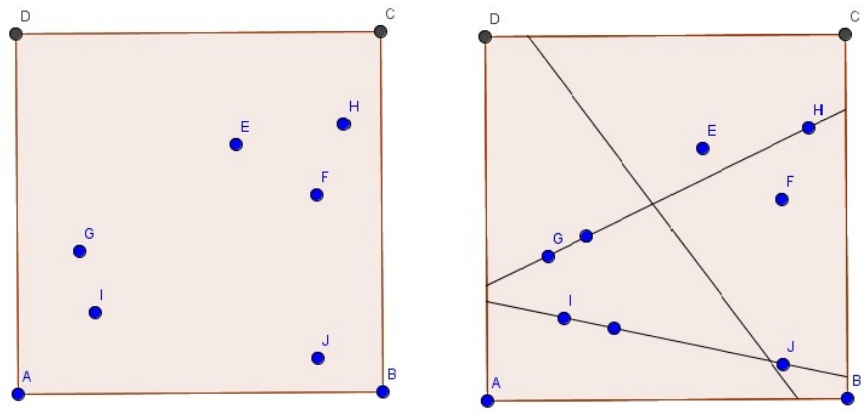

Fig. 6: Axiom 6.

The points without notations on the line $L(G,H)$ and $L(I,J)$ are the reflections about the fold line of the point E and respectively F.

The construction related to this axiom leads to compute the roots of a third order polynomial equation.

The reflection of the point $P(a,b)$ about the line $y = \lambda x + \mu$ has the coordinates $(\frac{1+2b\lambda-a\lambda^2-2\lambda\mu}{1+\lambda^2}, \frac{-b+2a\lambda+b\lambda^2+2\mu}{1+\lambda^2})$.

If $P_i(x_i, y_i)$ are two points and $y = m_i x + n_i$ are the equations of the lines L_i, $i = 1,2$, then the conditions that the fold $y = mx + n$ place P_i on the line L_i is

$$-n_i - \frac{m_i(-2mn + x_1 - m^2 x_i + 2my_i)}{1+m^2} + \frac{2n + 2mx_i - y_i + m^2 y_i}{1+m^2} = 0,\ i = 1,2. \tag{1}$$

Eliminating n there is obtained the polynomial equation

$$p(m) = n_1 - n_2 + m_1 x_1 - m_2 x_2 + y_1 - y_2 + \tag{2}$$

$$+m(m_2n_1-m_1n_2-2x_1+m_1m_2x_1+2x_2-m_1m_2x_2+2m_1y_1+m_2y_1-m_1y_2-2m_2y_2)+$$
$$+m^2(n_1-n_2-m_1x_1-2m_2x_1+2m_1x_2+m_2x_2-y_1+2m_1m_2y_1+y_2-2m_1m_2y_2)+$$
$$+m^3(m_2n_1-m_1n_2-m_1m_2x_1+m_1m_2x_2-m_2y_1+m_1y_2)=0.$$

From (1), for $i=1$, it results that

$$n=\frac{(1+m^2)n_1+m_1x_1+y_1-m^2(m_1x_1+y_1)-2m(x_1-m_1y_1)}{2+2mm_1}.$$

Within GeoGebra, the root $m0$ of (2) is computed as

$$m0=x(Root[p]).$$

This fold is the reason that in origami it is possible to solve some problems which cannot be solved with the ruler and the compass.

Axiom 7

Given a point P and two concurrent lines L_1 and L_2, there is a fold placing P onto L_1 and perpendicular to L_2.

If E is a point and $L(F,G)$, $L(H,I)$ are two lines then the tool $H7[A,B,E,F,G,H,I]$ draws the fold placing E on $L(F,G)$ and perpendicular to $L(H,I)$, Fig. 7.

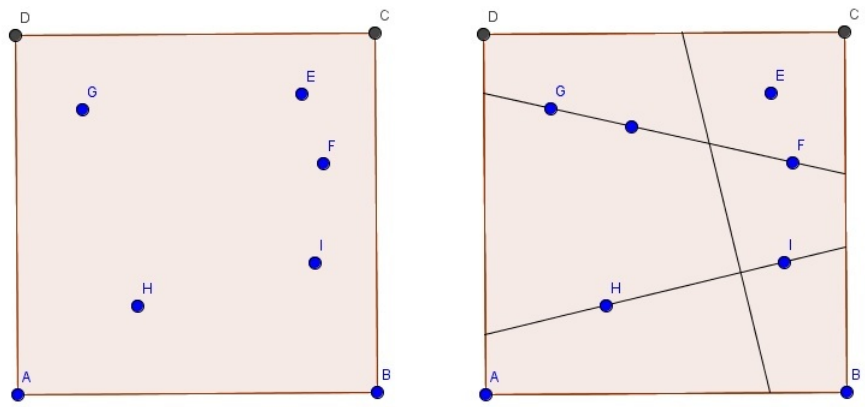

Fig. 7: Axiom 7.

Again, the point without notation on the line $L(F, G)$ is the reflection of the point E about the fold line.

Let J be the intersection between the parallel to $L(H, I)$ through E with $L(F, G)$. The fold line will be the perpendicular bisector of the the segment EJ.

Remark 2.1

Each tool use $H1$ as a procedure to draw only the segment of a line contained in the origami square. When a tools is saved it saves the $H1$ tool, too. If several tools are loaded then several instances of $H1$ are loaded.

Remark 2.2

GeoGebra is developed in Java programming language. The documentation of GeoGebra doesn't contains detailed data to create tools from user classes. Programming in Java would provide a more reliable variant than using the GeoGebra functions.

3 Examples of the use of the tools

At the beginning the tools must be loaded into GeoGebra (File/Open). We advise the reader to introduce the calling statement of a tools through the input line.

Example 3.1 *Construct a fold (line) parallel to a given line through any given point using origami.*

In the square $ABCD$, let be the point E and the line $L(F, G)$. The construction contains:

1. Call $H4[A, B, E, F, G]$
2. Select a point H on the generated line;
3. Call $H4[A, B, E, E, H]$

The final result is given in Fig. 8.

Example 3.2 *Construct an equilateral triangle with the base AB, the bottom side of the origami piece of paper, [5].*

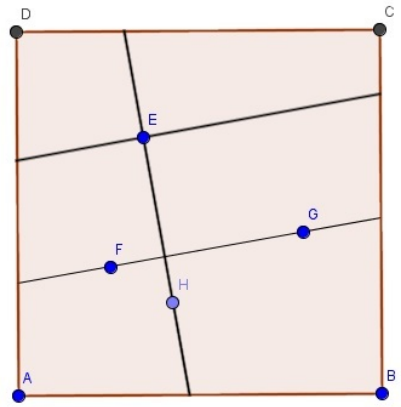

Fig. 8: Parallel line to $L(F, G)$ through the point E.

The sequence of the commands are:

1. Call $H3[A, B, A, D, B, C]$
2. Define F and G as the intersection points of the obtained fold line with the segments AB and respectively CD.
3. Call $H5[A, B, D, A, G, F]$
4. Define H the intersection between the obtained line with FG.
5. Call $H1[A, B, A, H]$
6. Call $H1[A, B, B, H]$

The final result is given in Fig. 9.

Example 3.3 *The trisection of an acute angle [3], [6].*

The construction is:

1. Select a point E on the segment AD and a point F on BC.
2. Call $H1[A, B, A, F]$

   The angle to be trisect will be $\widehat{BAF}$.
3. Call $H4[A, B, E, A, D]$
4. Call $H2[A, B, A, E]$

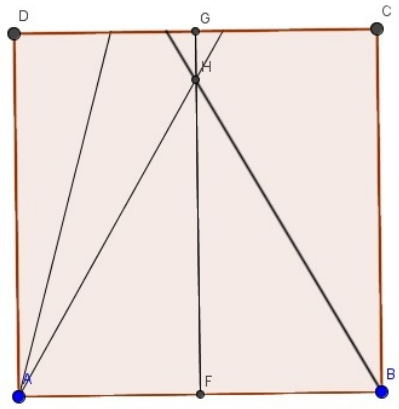

Fig. 9: Construction of the equilateral triangle.

5. Denote by H and I the intersections of the generated line with AD and BC.
6. Call $H6[A, B, A, E, H, I, A, F]$
7. Let M be the intersection between the last generated line with HI.
8. Call $H1[A, B, A, M]$

   Then $\widehat{MAF} = \frac{1}{3}\widehat{BAF}$, see [8] for the proof.

The final result is given in Fig. 10.

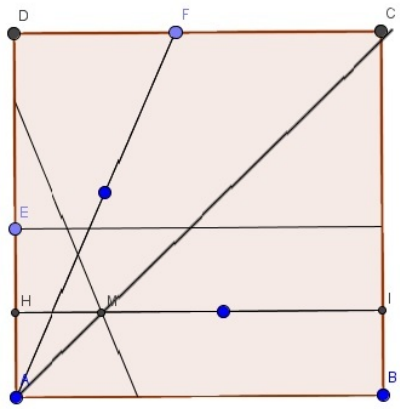

Fig. 10: The trisection of an acute angle .

References

[1] Alperin R.C., 2000, *A Mathematical Theory of Origami Constructions and Numbers.* New York Journal of Mathematics, **6**, 119-133.

[2] Auckly D., Cleveland J., 2004, *Totally real origami and impossible paper folding.* arXiv:math/0407174.

[3] Dacorogna B., Marcellini P., Paolini E., 2010, *Origami and Partial Differential Equations.* Notices Amer. Math. Soc., **57**, no. 5, 598-606.

[4] Hatori K., Origami versus Straight-Edge-and-Compass. `http://origami.ousaan.com/index.html`, K's Origami → Fractional Library → Origami Construction.

[5] Sundara Row T., 1917, *Geometric Exercises in Paper Folding.* The Open Court Publishing Company, Chicago, `http://www.arvindguptatoys.com/arvindgupta/sundararow.pdf`.

[6] Yin S., 2009, *The Mathematics of Origami.* `https://www.math.washington.edu/~morrow/336_09/papers/Sheri.pdf`

[7] * * *, `www.geogebra.org`.

[8] * * *, Trisecting the angle using origami – proof. `https://plus.maths.org/content/os/issue53/features/newton/angle`.